Rent or Toxicity? Strategic Exploitability of Learning Market Makers with Public Wallet Information (Hyperliquid)

Revised proposal and Workshop-1 report

Deepu Pradeep Miguel Betran Menz
6.S974 Games, Learning, and Security — Example 4 (Algorithmic Market with Adaptive Agents)

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Abstract

Independent reinforcement-learning market makers settle at spreads above the competitive level without communicating. A wide spread is either *rent* that an entrant can capture or the correct price of *toxic* order flow, and the two look identical on the tape. We study the entrant's problem on Hyperliquid, an on-chain order book where every print carries both counterparties' wallet addresses. This document revises our October 6 proposal in light of three days of simulation and the first Hyperliquid data. Three things changed. (1) The exploitability hypothesis "more collusive ⇒ more exploitable" holds only in a trivial form; the fraction of rent a profit-maximising entrant destroys is flat at ≈ 0.7 across populations, so RQ2 is restated as a measurement. (2) The mechanism by which wallet identity pays is not wallet persistence (same-wallet repeats are 3–4× chance but only 0.02–0.05 in level); it is that informed flow clusters in time and a toxic wallet's print reveals the regime *before the price does*. The simulator now has a latent toxicity regime and a lag between a print and its mark, and with them identity's value reappears where the data say it should. (3) The quoted spread on BTC, ETH and SOL is pinned at one tick essentially all the time, so on those books rent cannot live in the spread; it can only live in which fills a maker gets. A fill at the touch pays that one tick and loses 0.3–0.6 bps to the mid within ten seconds on every coin; the taker's cost is in depth—one order in ten walks the book and pays twenty ticks, and on the deep books those are the informed ones. We lay out the revised model, the results that forced the revisions, the calibration numbers, and the plan to a December preprint.

1 Motivation and central claim

Market makers (MMs) post bids and asks, earn the spread from impatient traders, and lose to *informed* traders whose orders predict the next price move. Increasingly they are learning algorithms. Three facts make this a trading-firm problem rather than only a regulator's. (i) Independent learning MMs drift to spreads above the competitive level without communicating [7, 1]. (ii) A spread above competitive is rent an entrant can capture by undercutting, but the same wide spread is also the correct response to toxic flow: identical observable, opposite action. (iii) Optiver reports that current LLM traders name adverse selection yet trade at negative expected value and fail to model how others react [16]. The skill a desk wants—read the tape, decide whether competitors are soft or the flow is poisonous, and quote accordingly without becoming the exploited party—has no quantitative benchmark.

Central claim, revised. Learning market makers earn supra-competitive rents that survive entry: a simple entrant that reads markouts off the public tape takes most of the rent and keeps it when the incumbents are allowed to re-learn, because the cartel re-forms around the entrant rather than collapsing. The entrant's problem is therefore not whether rent exists but *which flow to take*, and that is a regime-inference problem. On a venue with public wallet identity, the identity of the last taker reveals the toxicity regime before the price move that would reveal it anonymously; the lag between a print and its mark is where the value of identity lives. On tick-constrained books, where the spread cannot widen, the same fill-selection mechanism is the only place rent can live at all.

The pieces exist separately: learned rents [1, 7]; strategic exploitation of learners in abstract games [10, 4]; on Hyperliquid, persistent wallet histories carry adverse-selection information [2] and disclosed flow changes impact [3]. What is new here is the chain: learned rent → entrant best response → value of public identity, measured in the same units in the simulator and on the venue.

2 What changed since Deliverable 1, and why

Table 1 is the audit trail. Every row is a claim in the October 6 text, the finding that bears on it, and the change made. Code, results and a running findings log are at `https://github.com/pradeepu191/wargame` (`results/REPLICATION_NOTES.md`).

3 Model

3.1 Market

One contract, discrete time $t = 1, \ldots, T$ (one block per period, $T = 100$ per episode), N quoting MMs on a tick grid of size θ. Each MM i posts symmetric quotes at half-spread $h_t^i \in \{1, \ldots, K\}$ ticks around the mid m_t. One taker arrives per period: informed with probability α, else a liquidity trader with private valuation $m_t + L$, $L \sim N(0, \sigma_L^2)$, who buys if the best ask is below her valuation and sells if the best bid is above it. The fundamental is $V_t = m_t \pm d$, $d \sim \text{Exp}(\mu)$, redrawn each period; an informed taker trades only when d exceeds the best half-spread, so no quote is immune and, by memorylessness, an informed fill costs the maker exactly μ in expectation at every spread. (Deliverable 1's binary-V specification had an adverse-selection cliff—any half-spread beyond the edge was immune—which tabular learners find and sit on; the exponential edge removes it while keeping closed forms.) The taker hits the best quote; ties split uniformly; the maker's profit per fill is $(p - V_t)$ on a sale, $(V_t - p)$ on a purchase.

Benchmarks (closed form). With $x = h\theta$, the market-wide expected one-period profit of quoting h is

$$\Pi(h; \alpha) = (1 - \alpha)\, 2x\, \bar{\Phi}(x/\sigma_L) \;-\; \alpha\, \mu\, e^{-x/\mu},$$

linear in α. The competitive half-spread h^C is the first grid point with $\Pi \geq 0$ (Bertrand undercutting stops there), the joint-monopoly h^M maximises Π, and Π^C, Π^M are the profits at those points. The collusion index of a population is $\Delta = (\bar{\Pi} - \Pi^C)/(\Pi^M - \Pi^C)$ [11]; we write $C(\pi)$ for the same quantity when the population is frozen and evaluated. With $\theta = 0.5$, $K = 24$, $\sigma_L = 8$, $\mu = 5$: $(h^C, h^M) = (2, 13)$ at $\alpha = 0.1$, $(4, 14)$ at 0.3, $(8, 17)$ at 0.5; $\Pi^M - \Pi^C$ falls from 1.90 to 0.66, so less rent is available where flow is more toxic.

Deliverable 1 said	What we found	Change
RQ2 hypothesis $\partial L_{\mathrm{BR}}/\partial C > 0$: the regularity that sustains rents makes policies easier to manipulate.	Slope of L_{BR} on C is ≈ 1 by construction (more rent, more to lose); the *fraction* of rent destroyed is ≈ 0.7 with at most a weak dependence on C, 180 populations (§5.2).	Restated as a measurement; attacker set split into damage-maximising and profit-maximising.
Identity matters because wallet toxicity is a persistent trait (Zhai's ten-day persistence).	Same-wallet repeat rate at the order level is 0.02–0.05; identity's effect on the next fill's markout is 0.5–0.9 bps and survives conditioning on the anonymous tape (§5.5).	Simulator: latent toxicity regime + mark lag replace wallet persistence (§3.2); RQ1 restated.
Rent appears as a quoted spread above break-even; undercutting means quoting tighter.	Quoted spread on BTC/ETH/SOL is one tick, p90 = median, over six hours of book data (§5.5.1).	Spread game kept as theory and for books that open beyond a tick; on tick-bound books rent is reinterpreted as fill selection; queue model added to plan.
Markouts from the trade tape.	Trade-price proxy carries bid–ask bounce. Against the true mid the 10-s markout is negative on all four coins in both book windows, and negative for fills at the touch (-0.27 to -0.56 bps); the taker's cost is in depth, not the one-tick quote (§5.5.1).	Top-of-book recorded on every change; all markouts against the pre-trade mid.
MM wallets avoid each other (CCG signature).	Avoidance ratio ≈ 1 on 374 classified wallets; below 1 only among the top-10 on thin books; unstable across windows.	Not claimed until a fixed wallet set over a longer window exists.
Entrant quote rule: undercut when rent ≥ 2 ticks.	A margin makes a *better-informed* entrant earn less (it declines positive-EV fills when pessimistic).	Break-even rule (margin 1); the first identity grid re-run under it gives the same result (§5.4).

Table 1: Audit of claims in Deliverable 1 against findings to date.

3.2 Takers with identity: types, regime, and the mark lag

This subsection replaces the "persistent wallet id" sentence of Deliverable 1 and is the main modelling change.

Types. A population of M wallets ($M = 50$) with persistent types α_j, taken as stratified quantiles of $\mathrm{Beta}(\bar{\alpha}\kappa_\alpha, (1-\bar{\alpha})\kappa_\alpha)$ so that the mean is pinned at $\bar{\alpha}$ and the concentration κ_α sets only the dispersion ($\kappa_\alpha \to \infty$ recovers the homogeneous model; the subscript keeps it apart from the fraction of rent destroyed κ_A). Wallet j's order is informed with probability α_j.

Regime. A latent state $z_t \in \{\mathrm{calm}, \mathrm{toxic}\}$, kept with probability ρ_z and otherwise redrawn from a stationary law chosen so the stationary mean toxicity equals $\bar{\alpha}$. Wallet activity depends on the state: in the toxic state arrival weights are $\propto w_j\alpha_j$, in the calm state $\propto w_j(1-\alpha_j)$. With $\kappa_\alpha = 1$ the two states have toxicity 0.15 and 0.65 around $\bar{\alpha} = 0.3$. The identity of the last print is therefore

evidence about the regime the *next* arrival shares, whoever it is. (A separate same-wallet persistence parameter ρ remains in the code; the data put it near zero and it is off in all results below.)

Public information and the mark lag. After commitment, a print is public: event, price, and wallet id. Its *mark*—the ex-post value V_t against which the fill's profit is measured—becomes public L periods later (`mark_lag`). Deliverable 1 implicitly had $L = 1$, which gives an anonymous observer oracle-quality markouts before the next arrival. On Hyperliquid the next order arrives in 0.4–1.8 s and markouts realise over seconds to minutes; L is the number of arrivals that elapse before the tape reveals what identity revealed at the print. Information sets are nested:

$$\mathcal{F}_t^{\text{anon}} \subset \mathcal{F}_t^{\text{id}} \subset \mathcal{F}_t^{\text{oracle}},$$

no ids / committed wallet ids / the true types. Nobody ever sees the identity of the order about to hit them.

3.3 Agents

Incumbents. Tabular Q-learning [11, 1]: state = (own last half-spread, min rival last half-spread), action = half-spread, ε-greedy with $\varepsilon_t = e^{-\beta t}$, learning rate η, discount 0.95, 2×10^4 episodes. Incumbents are identity-blind by construction, so the identity experiments measure what identity is worth to a newcomer against an identity-blind cartel. "Re-learning" incumbents resume Q-learning after entry with a fresh exploration schedule.

Entrants. All see only public information.

- *Competitive-GM*: quotes $h^C(\bar{\alpha})$, ignores everyone. The textbook benchmark.
- *One-tick undercut*: one tick inside the best incumbent quote. The myopic upper bound if nobody punishes.
- *Own-fill markout* (Deliverable 1's "markout rule"): estimates toxicity from the sign of its own fills' profits and undercuts when the implied rent is positive.
- *Wallet entrant* under $\mathcal{F}^{\text{anon}}$, $\mathcal{F}^{\text{id}}$ or $\mathcal{F}^{\text{oracle}}$: exact grid-Bayes estimation of $\bar{\alpha}$ from every period (print or not, since the best quote is public) and of each wallet's α_j from its prints with a print-conditional likelihood (no fill-selection bias; after 300 episodes the population estimate is within 0.005 of truth and per-wallet error 0.03). A two-state filter tracks $b_t = P(z_t = \text{toxic})$: with ids the evidence at a print is *who* printed (likelihood ratio $\alpha_j(1-\bar{\alpha})/((1-\alpha_j)\bar{\alpha})$); without ids the evidence at the print is only that a print occurred, and the sign of its profit arrives L periods later and is carried to the present through ρ_z^L (the keep-or-redraw chain is reversible, so this is exact). Forecast $\alpha_{\text{next}} = b\,\alpha(\text{toxic}) + (1-b)\,\alpha(\text{calm})$. Quote rule: with best incumbent quote b^*, undercut to $\min(b^* - 1,\, h^M(\alpha_{\text{next}}))$ when $b^* - 1 \geq h^C(\alpha_{\text{next}})$, else quote $h^C(\alpha_{\text{next}})$, which for a toxic forecast is wider than the incumbents and amounts to withdrawing.

3.4 Statistics

- Δ / $C(\pi)$ as above; a *mechanism test*: in states where a rival quotes at or inside the monopoly point, is the learner's greedy action its myopic best response (undercut) or restraint? Restraint share, forgone one-period reward, and Q-gap are reported.
- Exploitability: Π_{nominal} (frozen incumbents alone), $J(\pi; A)$ (incumbents' profit with attacker A), $L_{\text{BR}}^A = \Pi_{\text{nominal}} - J$, $L_{\text{BR}} = \max_A L_{\text{BR}}^A$, and $\kappa_A = L_{\text{BR}}^A/(\Pi_{\text{nominal}} - \Pi^C)$, the fraction of rent destroyed. Entrant profit is reported in rent units $(\Pi^M - \Pi^C)$ so N and α are comparable.

- Value of information: $\pi^{\text{id}} - \pi^{\text{anon}}$ and $\pi^{\text{oracle}} - \pi^{\text{id}}$, paired by seed and market realisation (common random numbers).
- Adverse-selection shift: informed share of the entrant's fills vs the incumbents' fills vs the market without the entrant.

3.5 What is deliberately simple, and what the data say about it

Unit size, symmetric quotes, risk-neutral makers, exogenous taker mix, no inventory in the state (inventory buckets are implemented and off), no fees or rebates, no request budgets, no liquidation flow. Two of these are now known to matter on the venue. The spread on the major books is one tick (§5.5.1), so the symmetric-half-spread action does not describe BTC/ETH/SOL under normal conditions; it describes thin books (HYPE opens to two ticks) and stress. And the archive records each counterparty's pre-trade position on every print, so inventory paths of real MM wallets are observable and the inventory-skew benchmark can be calibrated rather than assumed.

4 Hyperliquid: mechanics and data

Mechanics as treatment arms (unchanged from Deliverable 1 except where noted). Persistent pseudonymous wallet addresses on every committed order and both sides of every print. Per-user request budgets earned by filling, so the only "drain" attack is *induced self-exhaustion* of a reactive rival. A priority fee that buys ≈ 45 ms/bp for aggressive orders and queue position for resting ones [15]. A price grid of at most five significant figures: at current prices one tick is \$1 on BTC, \$0.10 on ETH, \$0.01 on SOL, \$0.001 on HYPE, i.e. 0.12, 0.39, 0.87 and 0.11 bps respectively—and §5.5.1 shows the spread sits on it. A protocol vault (HLP) with undisclosed strategies, which we treat as a calibration target for a fixed-policy benchmark, not a known policy. Liquidation flow routed to the book, with the October 10, 2025 cascade as the stress case.

Archive. Albers et al. [14], Zenodo record 18184441: trades for all perpetuals October 2025–January 2026 (JSON lines, both counterparties with pre-trade position, order ids, TWAP flags); order lifecycles and book diffs for BTC/ETH/SOL in December 2025. Two facts not in the schema document that matter for use: `side_info` is [buyer, seller] (verified on 3.4M trades by the order-id rule: the aggressor's order is the newer one in 97.1% of trades; the other 2.9% are aggressors *older* than the resting order, which is what triggered stop and liquidation orders look like, and are flagged); and the L1 hash is all zeros for orders filled in their submission block, so it cannot be an order key—fills of one sweep are collapsed on (block time, taker, side). January coverage is Jan 1–3 and Jan 19–31; we used Jan 26–28 (72 h, 3.42M prints on BTC/ETH/SOL/HYPE, 1.44M taker orders).

Live feed. A recorder on the public WebSocket (`data/record.py`) writes trades with both wallets, L2 snapshots, and the top of book on every change. 24 h of trades (Oct 6–7, 871k prints) and 6 h with the book (Oct 8). The feed and the archive agree on every structural statistic (§5.5).

Pipeline. `analysis/calibrate.py` turns either source into one row per coin: activity, markouts (proxy and true-mid), same-taker repeat rate at the order level against its random-matching baseline, split-half rank correlation of per-taker markouts, the conditional markout after a toxic vs a benign wallet's order, the same split conditioned on the anonymous "price already moved against the last

maker" signal, MM wallet classification with a shuffled-label null for turn-taking, and the book statistics. Classification always uses the first half of a sample and measurement the second.

5 Results to date

All simulator results: $\theta = 0.5$, $K = 24$, $\sigma_L = 8$, $\mu = 5$, common random numbers across policies within a cell, means over seeds with seed-level standard deviations where quoted.

5.1 Replication of learned supra-competitive spreads

Two or three Q-learners ($\eta = 0.15$, $\beta = 4 \times 10^{-6}$, 10 seeds) settle at half-spreads of 10.6, 14.4 and 17.4 ticks for $\alpha = 0.1, 0.3, 0.5$ against competitive 2, 4, 8 and monopoly 13, 14, 17: $\Delta = 0.83$, 0.86, 0.77 (seed sd 0.01–0.02). A robustness grid at $\alpha = 0.3$ over $N \in \{2, 3, 5\}$, $\eta \in \{0.05, 0.15\}$ and optimistic/pessimistic initialisation gives $\Delta = 0.71$–0.87 in every cell (Table 2); the slow learners ($\eta = 0.05$) quote tighter. The mechanism test says this is restraint, not under-exploration: in 55–83% of visited states a rival sits at or inside the monopoly quote, and in 96–100% of those the learner's greedy action forgoes the myopic undercut. The best-quoter identity switches between episodes 45–70% of the time: a turn-taking convention, not a fixed leader.

N	η	init	half-spread	Δ	restraint | contested	leader switch
2	0.05	0	11.2	0.84	1.00	0.52
2	0.15	0	14.4	0.86	0.99	0.45
3	0.05	0	9.3	0.72	0.98	0.61
3	0.15	0	14.9	0.86	0.99	0.58
5	0.05	0	9.7	0.71	0.97	0.63
5	0.15	0	15.0	0.87	0.99	0.62

Table 2: Robustness grid at $\alpha = 0.3$ (10 seeds; optimistic-initialisation rows omitted, within 0.03 of these). $h^C = 4$, $h^M = 14$.

5.2 Does rent create vulnerability? (RQ2)

180 frozen populations (the robustness grid and the main runs at three α, ten seeds each; C from 0.42 to 0.97 at the seed level) each face four attackers for 500 episodes under common random numbers. Figure 1 and Table 3.

slope on C (OLS, bootstrap 95% CI)	pooled	$\alpha = 0.3$ only	within cell (seeds)
L_{BR}, max over attackers	1.02 [0.90, 1.18]	1.03 [1.00, 1.07]	1.07 [0.96, 1.22]
L_{BR}, markout attacker	0.71 [0.56, 0.90]	0.86 [0.71, 1.02]	0.74 [0.59, 0.98]
κ_A, markout attacker	−0.06 [−0.31, 0.26]	0.23 [0.03, 0.43]	−0.07 [−0.33, 0.31]
κ_A, undercut attacker	−0.17 [−0.71, 0.30]	0.25 [0.04, 0.49]	−0.30 [−0.94, 0.35]
κ_A, oracle attacker	−0.11 [−0.41, 0.22]	0.10 [−0.11, 0.34]	−0.07 [−0.46, 0.40]
entrant profit, markout attacker	0.34 [0.19, 0.54]	0.56 [0.38, 0.76]	0.15 [−0.01, 0.33]

Table 3: Regressions on rent across 180 frozen populations, ten seeds. κ_A is the fraction of rent destroyed. The five-seed version of this table (120 populations) is in the replication notes; every entry moved by less than its interval.

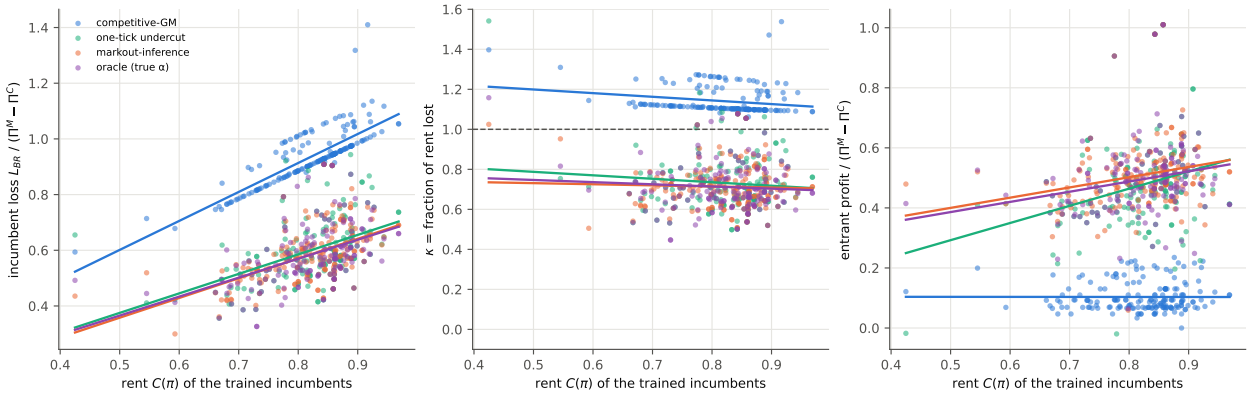


Figure 1: Best-response loss, fraction of rent destroyed, and attacker profit against the incumbents' rent $C(\pi)$, 180 populations (ten seeds), one point per population and attacker.

- The literal hypothesis holds, by construction. The attacker that minimises incumbent profit is the competitive-GM quoter in 179 of 180 populations, and it pushes the incumbents *below* Π^C (frozen incumbents sometimes respond to a quote at h^C by undercutting it and lose), so $L_{\text{BR}} \approx \Pi_{\text{nominal}}$ and the slope is one ($\kappa_A = 1.14$ on average, and slightly *decreasing* in C: -0.10 $[-0.14, -0.06]$ at $\alpha = 0.3$).
- The substantive version is not supported. κ_A, the fraction of rent a profit-maximising attacker destroys, is 0.71–0.73 for undercut, markout and oracle attackers across the whole range of C. Pooled and within-cell slopes include zero; at $\alpha = 0.3$ the OLS slope is $+0.23$ to $+0.25$ with intervals that just exclude zero while the rank correlation does not ($\rho = 0.08$–0.11, $p > 0.17$), which over the observed range of C is at most 0.13 in κ_A. Higher-rent populations are exploitable in proportion to their rent, with at most a weak excess.
- Damage and profit are different objectives. The attacker that hurts the incumbents most earns least (0.10 rent units); the ones that earn most (markout 0.51, oracle 0.49, undercut 0.48) leave the incumbents 27–30% of their rent. The markout rule is the attacker's own best response in 89 of 180 populations, the undercutter in 58, the oracle in 33.
- Knowing the true α adds nothing over estimating it from own fills (0.49 vs 0.51). In a stationary market the public tape carries the full value of the toxicity *level*. This is the observation that sent us to heterogeneous takers.

5.3 Entry: frozen and re-learning incumbents (RQ1, first form)

Against frozen two-maker cartels, the own-fill markout entrant captures 79%, 96% and 93% of the monopoly rent at $\alpha = 0.1, 0.3, 0.5$ (entrant profit 0.95, 0.72, 0.36 per period); the one-tick undercutter 84%, 89%, 75%; the competitive quoter 18%, 8%, 14%. The entrant wins 54–62% of fills, not all of them: it wins the fills at profitable prices.

With incumbents that resume learning ($\varepsilon_0 = 0.3$ decaying to 0 over 8000 episodes, Figure 2), the three policies produce three outcomes. The competitive entrant collapses the cartel: market $\Delta \to 0$, the best quote sits at h^C, the entrant wins every fill and earns exactly $\Pi(h^C)$, and the re-learning incumbents converge to zero profit. The undercut and markout entrants let the cartel *re-form around them*: Δ rises as exploration decays ($0.28 \to 0.62$ at $\alpha = 0.5$ for markout), the best quote drifts wider, and the entrant's final profit equals its frozen-incumbent profit (0.95/0.67/0.39 vs 0.95/0.72/0.36). Collapsing the cartel is the worst thing an entrant can do to itself: quoting at h^C leaves 59%, 82% and 72% of the markout entrant's profit on the table. Inference beats blind

undercutting at every α and every stage, most during exploration ($\alpha = 0.5$, first window: 0.25 vs 0.03), because the undercutter follows exploring incumbents into loss-making quotes.

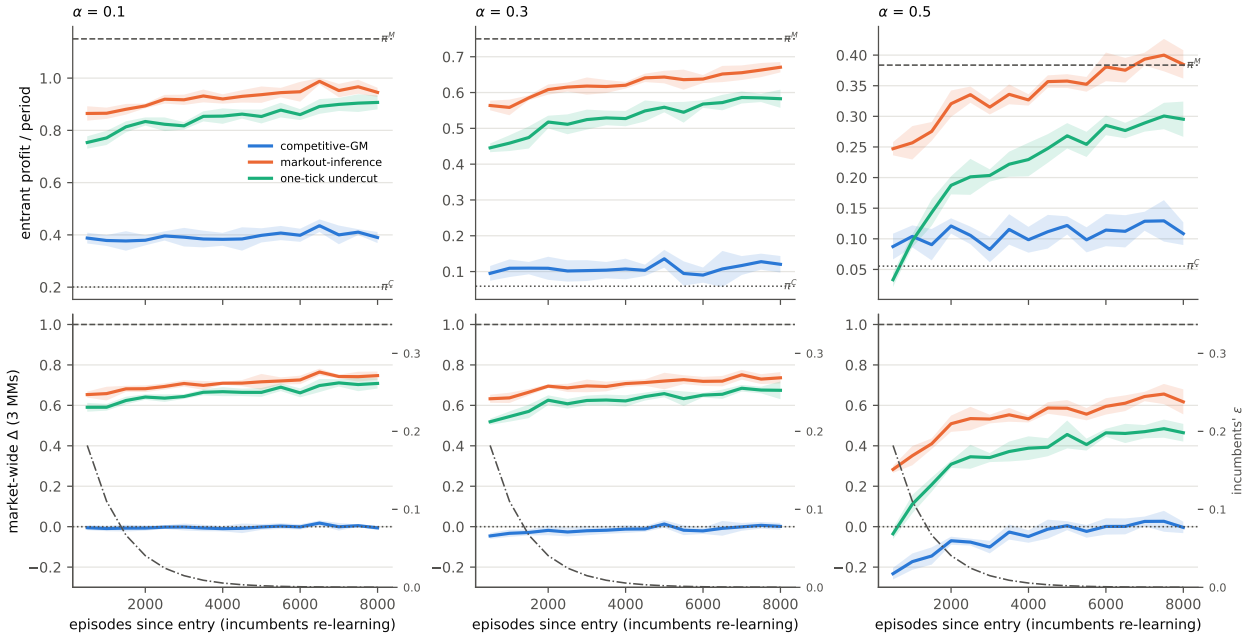


Figure 2: Entry against re-learning incumbents: entrant profit (top) and market-wide Δ (bottom) over 8000 episodes, incumbents' ε on the twin axis. Dashed/dotted lines are per-incumbent Π^M, Π^C of the pre-entry market.

5.4 The value of identity in the simulator (RQ1, revised form)

Three grids against frozen two-maker cartels, ten seeds (one trained cartel per seed), entrant profit in rent units, break-even quote rule. Differences between information sets are paired within a seed (same cartel, common random numbers) and reported with 95% intervals over seeds; the level of entrant profit varies by ±0.05–0.1 across seeds with the cartel the seed trained, so unpaired comparisons would hide everything below.

(a) Wallet persistence (the Deliverable-1 mechanism). With same-wallet persistence ρ and type concentration κ_α, identity is worth exactly nothing at $\rho = 0$ or $\kappa_\alpha = \infty$, and +0.15 to +0.30 rent units at $\rho = 0.9$–0.99, $\kappa_\alpha = 1$, $\bar{\alpha} \in \{0.3, 0.5\}$ (intervals ±0.05 to ±0.14; ten of ten seeds positive at $\rho = 0.9$); at $\bar{\alpha} = 0.1$ it is +0.03 to +0.07 and not distinguishable from zero. The identity entrant attains the oracle (oracle − id within ±0.04). The mechanism is fill selection: at $\bar{\alpha} = 0.5$, $\rho = 0.99$ it withdraws in 31% of periods, its fills are 18% informed against the incumbents' 53%, and the identity-blind cartel's residual book goes below Π^C while market Δ stays at 0.9—the rent moved to the entrant (Figure 3). The data put ρ near 0.03, so this channel is the theory check, not the mechanism.

(b) Regime, mark revealed immediately ($L = 1$). Replacing wallet persistence by the regime (§3.2) with $\rho = 0$, identity's increment is $+0.03 \pm 0.02$ at $\bar{\alpha} = 0.3$ and $+0.06 \pm 0.02$ at $\bar{\alpha} = 0.5$ ($\kappa_\alpha = 1$), and zero at $\bar{\alpha} = 0.1$ and at $\kappa_\alpha = 5$: the anonymous tape already carries the regime,

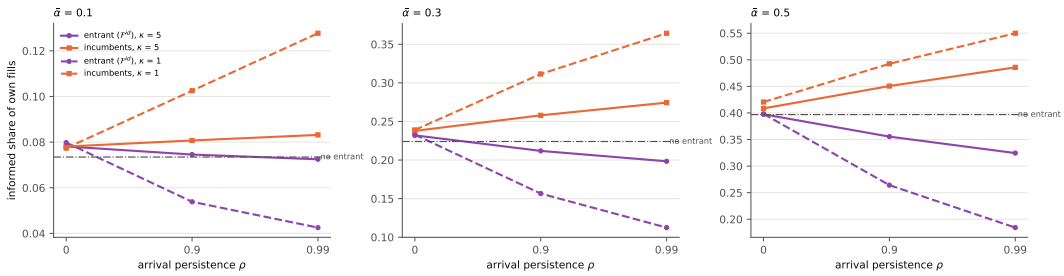


Figure 3: Adverse-selection shift under wallet persistence: informed share of the identity entrant's fills vs the incumbents', by persistence ρ and concentration κ_α; ten seeds.

because the last print's *outcome* is as good a regime signal as its author, and identity pays only where the toxic state crosses the entrant's decision threshold.

(c) Regime with the mark lag. Table 4 and Figure 4. At $\bar{\alpha} = 0.5$ the anonymous entrant's profit falls as the mark arrives later ($0.95 \to 0.84$ at $\rho_z = 0.99$) while the identity entrant's stays at 1.00–1.01, so identity's increment grows from +0.06 to +0.16 (± 0.07) at $\rho_z = 0.99$ and from +0.06 to +0.11 (± 0.05, peak at $L = 20$) at $\rho_z = 0.9$; nine or ten of ten seeds positive in every cell. The peak is informative: at $\rho_z = 0.9$ a regime lives about ten arrivals, and a mark that is sixty arrivals late is no worse than one twenty late because the tape was already useless at twenty. Identity's value grows with the mark lag as long as the lag is shorter than the regime's lifetime. At $\bar{\alpha} = 0.3$ the increment is +0.02 to +0.04, distinguishable from zero only at $\rho_z = 0.99$ and $L \geq 20$ (ten of ten at $L = 60$): against ~ 11-tick incumbents even the toxic state usually leaves an undercut profitable, so the regime is rarely decision-relevant. At $\bar{\alpha} = 0.1$ it is zero or slightly negative. The value of the wallet tag to an inside quoter is largest where the resting spread is close to the toxic-regime break-even—a prediction about *which* books reward identity-aware quoting, and the reason the one-tick books of §5.5.1, where the spread cannot move at all, are where identity should matter most (§5.6).

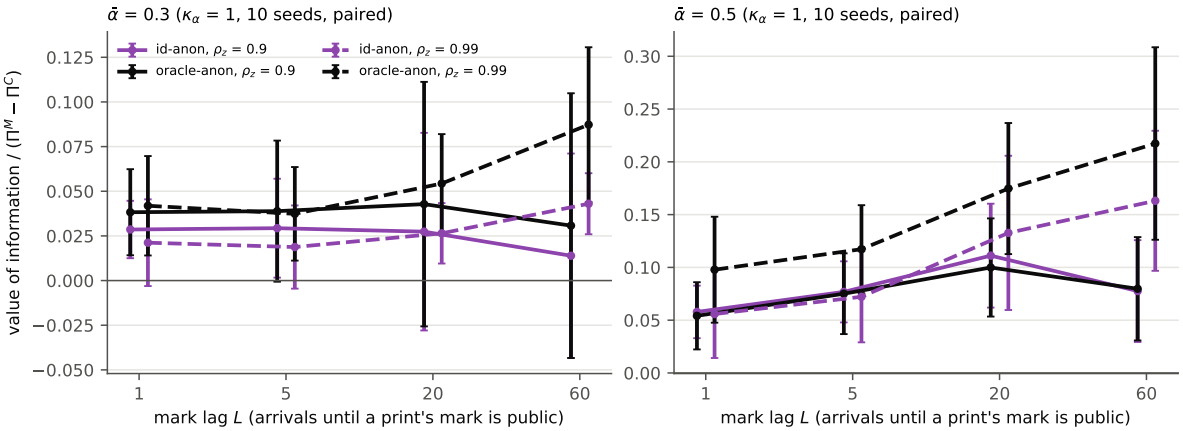


Figure 4: Value of identity (id − anon) and of the true regime (oracle − anon) in rent units against the mark lag L, paired per seed, $\kappa_\alpha = 1$, ten seeds, 95% intervals.

(d) Where information hurts. At $\kappa_\alpha = 5$ the types lie within 0.2 of each other, a wallet's identity carries little regime information, and the mean increment is zero—but the seed distribution has a tail: in three of ten cartels at $\bar{\alpha} = 0.3$, $\rho_z = 0.9$, $L = 20$ the anonymous entrant out-earns both

$\bar{\alpha}$	ρ_z	anon ($L = 1/5/20/60$)	id	oracle	id − anon, paired (±95%)
0.3	0.9	0.65 / 0.65 / 0.64 / 0.66	0.68 / 0.68 / 0.67 / 0.67	0.69	$+0.03 \pm .02$ / $+0.03 \pm .03$ / $+0.03 \pm .06$ / $+0.01 \pm .06$
0.3	0.99	0.76 / 0.77 / 0.75 / 0.72	0.78 / 0.78 / 0.78 / 0.76	0.80	$+0.02 \pm .02$ / $+0.02 \pm .02$ / $+0.03 \pm .02$ / $+0.04 \pm .02$
0.5	0.9	0.71 / 0.69 / 0.66 / 0.68	0.76 / 0.76 / 0.77 / 0.76	0.76	$+0.06 \pm .02$ / $+0.08 \pm .03$ / $+\mathbf{0.11} \pm \mathbf{.05}$ / $+0.08 \pm .05$
0.5	0.99	0.95 / 0.94 / 0.88 / 0.84	1.01 / 1.01 / 1.01 / 1.00	1.05	$+0.06 \pm .04$ / $+0.07 \pm .04$ / $+0.13 \pm .07$ / $+\mathbf{0.16} \pm \mathbf{.07}$

Table 4: Entrant profit in rent units under the regime model with mark lag L ($\kappa_\alpha = 1$, $\rho = 0$, break-even quote rule, ten seeds).

the identity and the oracle entrant by 0.2–0.5 rent units. In those cartels the anonymous entrant's steady undercut takes 86–99% of fills and the incumbents earn nothing; the better-informed entrants' quotes move with their regime belief, and the incumbents' learned reaction takes back 26–33% of the fills. An entrant that uses information naively can provoke the defenders. The reaction function is part of the game—the steering channel of RQ3, met here by accident—and the reason every identity number in this document is reported with its minimum over seeds.

5.5 Hyperliquid data

Table 5 gives the archive sample and the live sample side by side.

archive Jan 26–28 / live Oct 6–8	BTC	ETH	HYPE	SOL
orders / min	97 / 84	68 / 49	135 / 59	34 / 25
same-taker repeat (order level)	0.020 / 0.021	0.034 / 0.034	0.038 / 0.048	0.045 / 0.049
random-matching baseline	0.005 / 0.004	0.009 / 0.006	0.012 / 0.013	0.011 / 0.014
same-taker repeat (fill level; sweeps)	0.57 / 0.49	0.53 / 0.45	0.53 / 0.64	0.43 / 0.47
type rank corr., split-half	0.17 / 0.26	0.39 / 0.23	0.11 / 0.28	0.42 / 0.58
maker markout 10 s, proxy (bps)	−0.69 / −0.15	−0.79 / −0.83	−0.12 / −0.42	−0.41 / −0.40
next fill's markout after toxic / benign (bps)	−0.90 / −0.09	−0.77 / −0.21	−0.73 / +0.11	−0.84 / +0.04
identity alone (bps)	−0.81	−0.55	−0.84	−0.88
anonymous signal alone (bps)	−0.74	−0.55	−0.31	−0.46
identity given the anonymous signal (bps)	**−0.63**	**−0.47**	**−0.74**	**−0.87**
P(adverse move before next order)	0.18	0.21	0.21	0.20

Table 5: Calibration statistics. Archive rows from Jan 26–28 (72 h); live rows from the Oct 6–8 recorder (36 h). Identity rows are from the archive sample, second half, toxic wallets classified on the first half.

- **Feed and archive agree.** Rates, sweep collapsing, repeat rates, type persistence and the identity split reproduce; the market state differs (BTC was far less toxic in October), as it should.
- **Toxicity is a persistent wallet type** (rank correlations 0.1–0.6, all $p < 0.002$) but wallets rarely return consecutively: the order-level repeat rate is 3–4× random matching and 0.02–0.05 in level. Wallet persistence is not the mechanism.
- **Identity predicts the next fill's toxicity beyond what the tape has revealed.** Conditioning on the previous taker's class moves the next fill's markout by 0.55–0.88 bps against a mean of −0.1 to −0.8; 78–99% of that survives conditioning on whether the price has already moved against the previous maker, which happens only one time in five by the next arrival. This is the regime-plus-lag mechanism of §3.2, measured.

5.5.1 The book

Two book windows: six hours on October 8 from the throttled `l2Book` feed (one snapshot per 5.4 s) and six hours on October 11–12 from the `bbo` feed (every top-of-book change; median gap

0.13 s). Two alignment rules were needed, both learned from the second window and both now in `analysis/l2.py`: a trade and the top-of-book update it causes carry the same block timestamp, so a fill's reference mid is the last snapshot *strictly before* the trade (at-or-before returns the post-trade book and gave a negative effective spread on BTC); and marks are computed only inside the recorder's coverage window. Table 6 is the second window; the first window's ten-second markouts are kept as a row for comparison.

The quote is one tick; the cost is in depth. The quoted spread is one tick on BTC, ETH and SOL in both windows (p90 = median). A fill at the pre-trade touch pays exactly the quoted spread—effective 0.12 / 0.42 / 0.17 / 0.92 bps against quoted 0.12 / 0.39 / 0.12 / 0.90—which is also the check that the alignment is right. But 20–46% of prints are not at the touch: one order in ten (one in five on HYPE) walks past it, and those prints pay 2.4–5.2 bps, twenty ticks on BTC. Print-weighted, the effective spread is 0.73 / 1.01 / 1.73 / 1.76 bps, six times the quoted spread on BTC and fifteen on HYPE; equal-weighted across orders (the order's volume-weighted price against the pre-trade mid) it is 0.27 / 0.60 / 0.40 / 1.09, 1.2–3.4× quoted. On a one-tick book the maker's price decision is not the inside quote but how much to show at the inside and behind it.

The ordinary fill loses, and it is not a sweep artefact. Against the fresh mid the maker is under water at one second on BTC, ETH and HYPE (-0.26 / -0.30 / -0.19 bps; SOL $+0.15$) and at ten seconds on all four (-0.52 / -0.92 / -0.25 / -0.02). The October 8 level—-0.30 to -0.36 on every coin—did not reproduce; the sign did, on every coin in both windows, and in both windows the adverse move exceeds the half-spread earned (0.06 / 0.20 / 0.06 / 0.45). Splitting by where the print landed settles whether large sweeps drive this: fills *at the touch*, the fill a quiet maker gets, lose -0.41 / -0.56 / -0.45 / -0.27 bps at ten seconds on all four coins. Walked fills split by book. On the deep BTC and ETH books the makers an order walks to lose more than the touch (-0.80, -2.26): an order that pays twenty ticks to walk BTC is informed and its impact is permanent. On HYPE and SOL walked fills break even or earn (-0.01, $+0.98$): walking a \$6k-deep or 0.9-bps-wide book is cheap and uninformative, the price reverts, and the maker resting deep keeps the walk. This is the regime contrast of §3.2 measured by depth instead of by wallet. The identity split against the true mid holds on BTC, HYPE and SOL (after toxic minus after benign -0.70 / -0.48 / -0.47 bps) and is $+0.06$ on ETH at six hours of marked fills; the 125.8-hour trades-only split (-0.66 / -1.02 / -0.52 / -0.63 given the anonymous signal) is the one to quote for identity.

Who earns the spread. Every marked fill carries the maker's wallet, so the RQ3 anchor can be read directly (Figure 5; the top-10 MM wallets are chosen by volume over the full 125-hour window, not on the six hours being scored; SEs cluster by five-minute block). The hierarchy the simulator predicts is in the data: makers the classifier does not call MMs lose -0.26 to -1.97 bps per fill at ten seconds, the other MMs -0.11 to -0.74, and the top-10 MMs break even or earn (-0.14 / $+0.09$ / $+0.04$ / $+0.58$ on BTC/ETH/HYPE/SOL, holding 9–23% of fills). Restricting to fills *at the touch*—the same price, the same tick—the top-10 earn 0.4–1.1 bps per fill more than everyone else (-0.04 / $+0.24$ / $+0.01$ / $+0.60$ against -0.45 / -0.84 / -0.60 / -0.54): makers resting at the same price get different fills, and the difference is *when* they are at the front. Part of the edge is not being swept (top-10 touch share 0.70–0.94 against 0.43–0.79). The skill is persistent: the split-half rank correlation of per-maker markout is 0.57–0.86 ($p < 10^{-3}$) on every coin. Three to eight makers per coin have a significantly positive realized spread and they take 7–22% of all fills; the rest of the book loses to the mid within ten seconds. Makers who select earn the spread; identity-blind resting liquidity is the counterparty of that rent—the selection cartel of §5.6, observed.

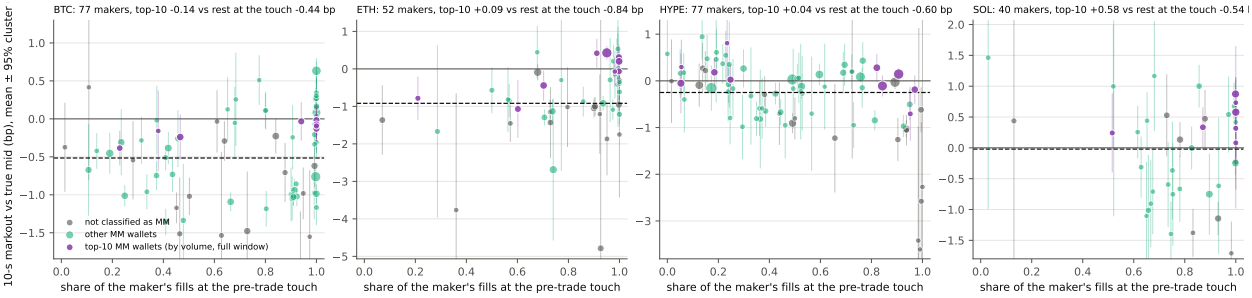


Figure 5: Per-maker 10-s markout against the true mid (bps, mean ± 95% cluster interval) by the share of the maker's fills at the pre-trade touch; one dot per maker with ≥ 100 marked fills, size by fills; dashed line: all fills. Oct 11–12 2026, six hours.

	BTC	ETH	HYPE	SOL
quoted spread, median / p90 (bps)	0.12 / 0.12	0.39 / 0.39	0.12 / 0.58	0.90 / 0.90
one tick (bps)	0.12	0.39	0.11	0.86
top-of-book depth, median (USD)	611k	283k	6.2k	78k
prints at the pre-trade touch	73%	79%	54%	80%
orders that walk past the touch	10%	10%	19%	8%
effective spread, fills at the touch (bps)	0.12	0.42	0.17	0.92
effective spread, walked fills (bps)	2.35	3.20	3.57	5.20
effective spread, print-weighted / per order (bps)	0.73 / 0.27	1.01 / 0.60	1.73 / 0.40	1.76 / 1.09
true-mid maker markout, 1 / 10 / 60s (bps)	−0.26 / −0.52 / −0.51	−0.30 / −0.92 / −0.38	−0.19 / −0.25 / −0.10	+0.15 / −0.02 / +0.05
10-s markout, at the touch / walked (bps)	−0.41 / −0.80	−0.56 / −2.26	−0.45 / −0.01	−0.27 / +0.98
10-s markout, Oct 8 window (bps)	−0.33	−0.33	−0.36	−0.30
after toxic / after benign, true mid (bps)	−0.97 / −0.27	−0.73 / −0.79	−0.65 / −0.17	−0.48 / −0.01

Table 6: Book statistics at top-of-book resolution, Oct 11–12 2026, six hours (41k / 26k / 47k / 15k marked fills). "At the touch": the print is at the best quote of the last book before the trade; "walked": beyond it.

5.6 The one-tick queue game (RQ3, first result)

Section 5.5.1 says the spread on the major books cannot move. We therefore built the game in which it does not: the spread is fixed at h ticks; each period every maker chooses to *rest* at the inside or *step out*; resting makers form a time-priority queue; the taker hits the front for one unit; the filled maker's order is consumed and it rejoins at the back if it keeps resting, so fills rotate among those who stay; a maker that steps out and returns joins at the back. Takers, wallet types, regime and mark lag are those of §3.2. The expected value of a fill at h against a taker of type α is $(1-q)x - q\mu$ with q the fill-conditional informed probability; at the population break-even $h = h^C$ it is +0.15 on average, +1.10 in the calm state and −2.26 in the toxic one ($\kappa_\alpha = 1$): all the value is in selection. One tick above break-even (+0.58 average) a coarse tick leaves the inside slightly supra-competitive and the queue shares that rent.

The incumbents are two always-resting makers—the tick-constrained competitive book—and the entrant is the wallet entrant with the decision "rest iff the fill value against the regime forecast is positive". Figure 6 and Table 7.

- At break-even the always-resting maker earns ≈ 0, as the BTC book data say the average resting fill does, and the identity entrant earns 0.14–0.16 per period by resting only when the regime is calm. Its fills are 16% informed against the incumbents' 29–31%, and the incumbents' profit goes to zero or below: the toxic flow is left to those who stay.
- The anonymous selector's edge collapses with the mark lag (+0.06 over always-resting at $L = 1$, +0.01 at $L = 20$, $\rho_z = 0.9$) while the identity entrant's does not (+0.10). A tape that is twenty arrivals stale cannot time the regime; identity can. The identity entrant attains 90–100% of the oracle.

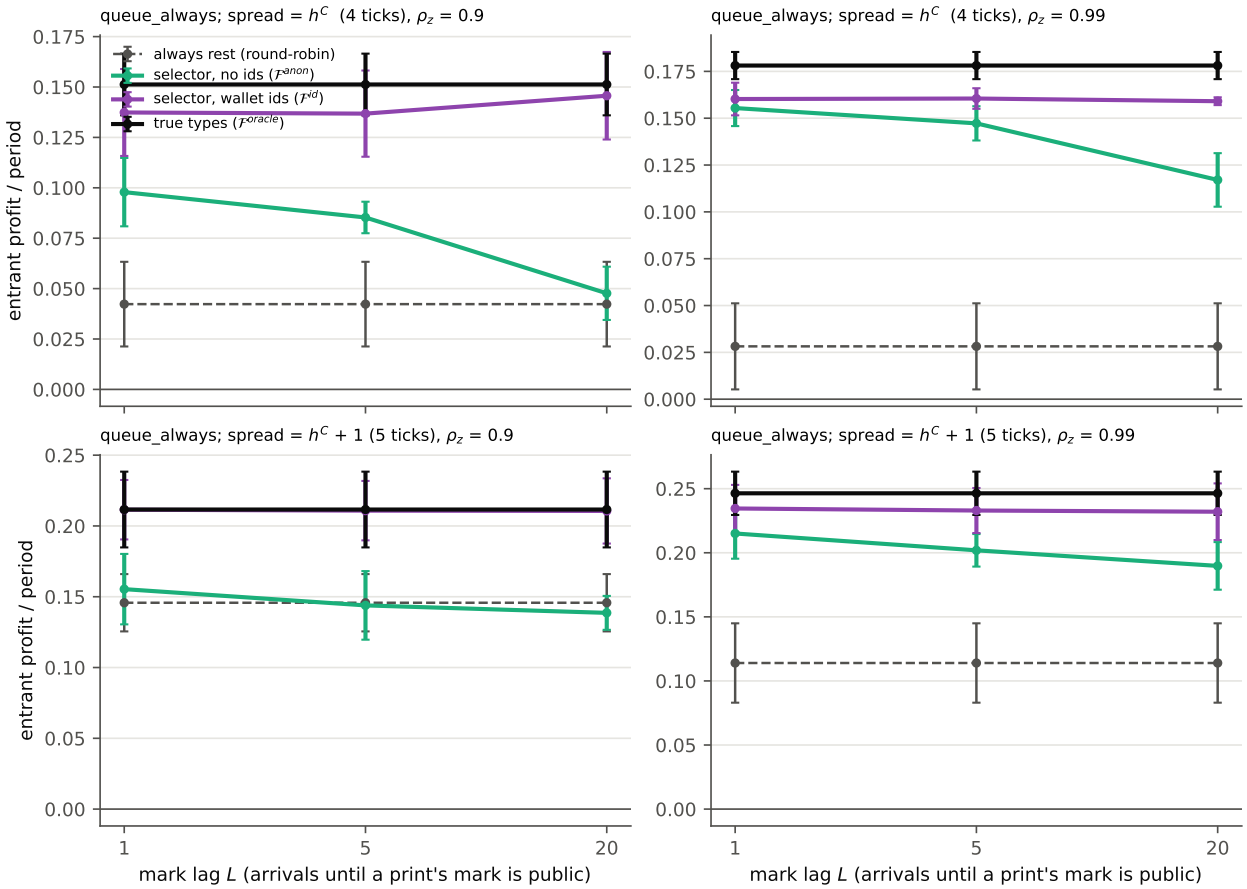


Figure 6: Queue game: entrant profit per period by information set against the mark lag, at break-even (top) and one tick above (bottom), $\rho_z = 0.9$ (left) and 0.99 (right); $\kappa_\alpha = 1$, $\bar{\alpha} = 0.3$, 5 seeds, 95% intervals.

- One tick above break-even the rotation shares the tick rent (0.15 each) and identity adds about 45% to it (0.21), taken from the incumbents ($0.14 \to 0.11$).

Learned incumbents. Two things had to change for incumbents that learn. First, a *fringe*: two always-resting makers in the population, the retail limit orders and the vault that never step out. Without them two learners that both step out in a toxic spell leave an empty book, no prints, a frozen tape signal, and a self-fulfilling lock in which nobody ever rests again. Second, the learner. Tabular Q-learning cannot learn to queue: with time priority the reward of resting arrives only after several consecutive periods in line, an ϵ-greedy learner breaks the chain, never reaches the front, sees zero whether it stays or leaves, and flickers (at 12,000 episodes it rests 40–60% of periods and holds 0–4% of fills; rank in the state and optimistic initialisation do not help). The proposal's other incumbent class does: a no-regret learner (EXP3) over complete queue policies {always rest, never rest, anonymous selector} converges in a few hundred episodes. Figure 7.

- *What the incumbents learn.* When the tape is fresh ($L \leq 5$, or $\rho_z = 0.99$ at any lag) both incumbents become anonymous selectors (expert weight 0.9–1.0); when it is stale ($L \geq 20$ at $\rho_z = 0.9$) a third to 40% of the weight moves to always-rest; one tick above break-even at $\rho_z = 0.9$ half of it does, because the rotation rent of the tick is worth more than selecting. Nobody learns to stay out. The learned incumbents earn 0.03–0.10 per period at break-even against 0.00–0.03 for the fringe, with 15–26% informed fills against the fringe's 26–31%: they select.

spread	ρ_z	L	always rest	anon	id	oracle	id − anon (paired)
h^C	0.9	1 / 5 / 20	0.04	0.10 / 0.09 / 0.05	0.14 / 0.14 / 0.15	0.15	+0.04 / +0.05 / +**0.10**
h^C	0.99	1 / 5 / 20	0.03	0.16 / 0.15 / 0.12	0.16 / 0.16 / 0.16	0.18	+0.01 / +0.01 / +0.04
$h^C + 1$	0.9	1 / 5 / 20	0.15	0.16 / 0.14 / 0.14	0.21 / 0.21 / 0.21	0.21	+0.06 / +0.07 / +0.07
$h^C + 1$	0.99	1 / 5 / 20	0.11	0.22 / 0.20 / 0.19	0.24 / 0.23 / 0.23	0.25	+0.02 / +0.03 / +0.04

Table 7: Queue game, entrant profit per period (price units); always-resting incumbents, 300 episodes, 5 seeds. Identity's increment is positive in every cell and in every seed but one.

- *Identity is worth more against incumbents that select, not less.* Paired over seeds, id − anon is +0.05 to +0.06 per period at break-even, $\rho_z = 0.9$, at every lag (five of five seeds; against always-resting incumbents with the same fringe it is +0.03 to +0.04). At $\rho_z = 0.99$ it grows with the lag from 0.00 to +0.06. The anonymous entrant's edge over always-resting shrinks with the lag (+0.04 → +0.01 at $\rho_z = 0.9$): once the incumbents read the tape, an entrant that reads only the tape is one more of them, and the wallet tag is what is left.
- *The fringe pays.* With an always-resting entrant the fringe earns +0.00 to +0.03 at break-even; with a selecting entrant, −0.02 to −0.04, and −0.10 at $\rho_z = 0.99$. Against no-regret incumbents at $\rho_z = 0.99$ the always-resting *entrant* is itself under water (−0.04 at $L = 1$). The identity-blind resting liquidity is the counterparty of the selection rent—the structure the per-maker data show (§5.5.1).

The spread game and the queue game give the same answer through different actions: a quoter who knows who just printed captures the benign flow and leaves the toxic flow to identity-blind rivals—by undercutting selectively where undercutting is possible, by resting selectively where it is not. The priority fee as the price of a place in line is the next step.

5.7 Synthesis

The chain the proposal set out to test now reads, with numbers: learning makers earn rent ($\Delta \approx 0.8$) that survives a best-responding entrant (the cartel re-forms around it; entrant keeps 79–96% of the monopoly rent); the entrant's binding problem is fill selection, i.e. reading the toxicity regime; on Hyperliquid the regime is readable from the wallet tag 0.5–0.9 bps before the price reveals it; and on the major books, where the spread cannot widen, fill selection is the only place rent can live. The queue game of §5.6 closes the loop in the simulator: on a book where the spread cannot move, the identity entrant earns the selection rent and the identity-blind makers absorb the toxic flow. The book adds the second action the queue model needs: a fill at the touch pays one tick, but one order in ten walks the book and pays twenty, and on BTC and ETH those are the informed ones—depth shown behind the inside is where the toxic flow lands. The same step on real quotes is now in (§5.5.1): the makers who select fills earn 0.4–1.1 bps per fill more than everyone else at the same price, persistently, while the average fill at the touch loses 0.3–0.6 bps within ten seconds. What remains is to say *how* they select—queue position, cancellation, depth, identity—which needs order-level data (the December L4 lifecycles) rather than prints.

6 Research questions, restated

RQ1 (Executable value of identity). Under $\mathcal{F}^{\text{anon}} \subset \mathcal{F}^{\text{id}} \subset \mathcal{F}^{\text{oracle}}$, what is an entrant's profit, and how does $\pi^{\text{id}} - \pi^{\text{anon}}$ depend on regime persistence ρ_z, type concentration κ_α, the mark lag L, and the incumbents' quote relative to the toxic-regime break-even? *Prediction (from §5.4)*: the increment is zero when the toxic state leaves an undercut profitable, and otherwise grows with L up

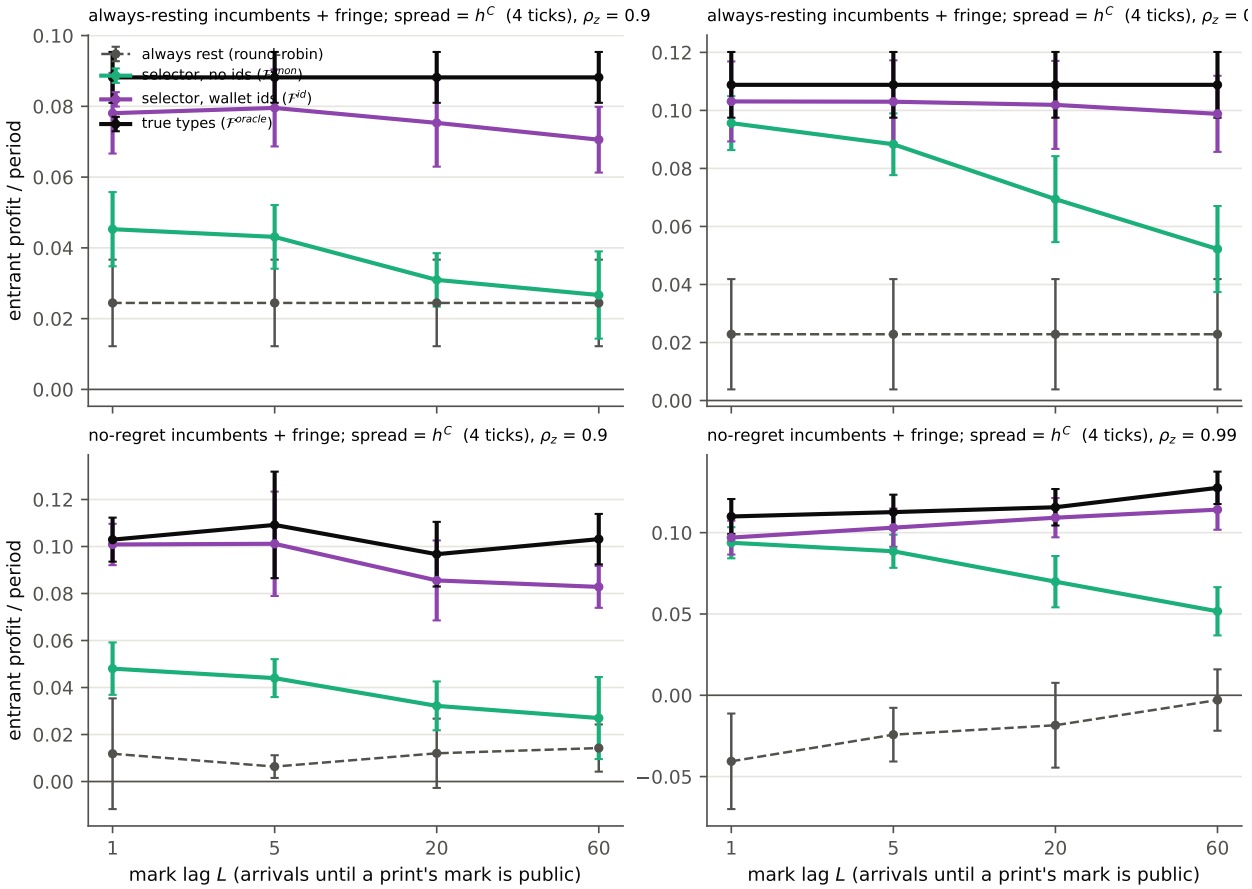


Figure 7: Queue game at break-even with a fringe of two always-resting makers: entrant profit per period by information set against the mark lag, against always-resting incumbents (top) and no-regret incumbents (bottom); $\rho_z = 0.9$ (left), 0.99 (right); $\kappa_\alpha = 1$, $\bar{\alpha} = 0.3$, 5 seeds, 95% intervals.

to the regime's lifetime. *Empirical counterpart*: the identity-given-anonymous-signal split by coin, against each coin's spread relative to break-even.

RQ2 (Exploitability, as measurement). For each incumbent population, report L_{BR} under the damage-maximising attacker and under the attacker's own best response, and κ_A. The claim supported so far: $\kappa_A \approx 0.7$ with at most a weak dependence on C (180 populations, ten seeds); a rational entrant captures about two thirds of what it destroys and the rest passes to takers as tighter quotes. Open: whether attackers that target the *reaction function* (steering, [10, 4]) rather than the quote level break the flatness of κ_A.

RQ3 (Tick-constrained books; new). On a one-tick book, model the maker's action as queue position and cancellation at the inside rather than spread width, with the priority fee as the price of position. Does the identity signal translate into realized spread for makers who use it, and is the resulting equilibrium a selection cartel—makers who leave the toxic flow to each other—rather than a spread cartel? *Results so far* (§5.6): yes—identity is worth +0.04 to +0.10 per period at break-even against always-resting makers and +0.05 to +0.06 against incumbents that learn to select on the tape, growing with the mark lag; the identity-blind resting liquidity absorbs the toxic flow in the simulator and on Hyperliquid (§5.5.1). Open: the priority fee; identity-aware incumbents.

Empirical anchor: per-wallet realized spread of the top makers vs the average resting fill, from the December L4 lifecycles.

RQ4 (Defences; extension). Quoter levers (randomisation, hysteresis, budget reserve, identity-aware control) and venue levers (anonymity, tick, request allowance, priority mechanism), reported as a profit–resilience frontier so "robust" cannot mean "withdrew".

7 Plan to December

1. *Now.* The October archive tar (contiguity, the cascade); a multi-day book run from an always-on machine, to put intervals on the touch-fill markout and the per-maker persistence by coin and hour.
2. *Late October.* Queue model for RQ3 extended with depth: makers choose how much to show at the inside and behind it, and cancel; time priority; priority fee; sweeps that walk the book as the informed event (§5.5.1). Calibrate L from inter-arrival vs markout horizon, ρ_z and the regime contrast from the conditional markouts, depth from the book.
3. *November.* Re-learning and identity-aware incumbents; a steering attacker; held-out attackers; inventory from `start_pos`; the December L4 lifecycles for reaction functions $P(\text{requote} \mid \text{rival undercuts})$ and for cancellation-before-toxic-flow.
4. *December.* Preprint. Rigor throughout: common random numbers, seed intervals, CVaR alongside means, train-at-α/test-at-α', chronological splits on data, classification on the first half only.

Feedback requested. (1) Given §5.5.1, is the queue-position game the right next model, or should we keep the spread game and restrict the empirical claims to books and episodes where the spread opens? (2) Is "rent as fill selection" a recognisable object in the market-microstructure literature we should cite, or a reframing we need to defend? (3) What venue would you consider the right second laboratory, given that the one-tick finding may be specific to Hyperliquid's five-significant-figure rule?

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